

PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	Scalability & Future Plan	DATE	July 2026

Scalability & Future Plan

Current Limitations

- Model trained on a single historical dataset; may need periodic retraining as applicant patterns evolve
- Class imbalance (~88% approved / 12% rejected) limits recall on the minority class
- Currently deployed as a local Flask development server, not a production-grade deployment

Future Enhancements

- Deploy the model via IBM Watson Machine Learning for scalable, cloud-hosted predictions
- Introduce additional engineered features (e.g., number of past-due months, number of loans) to align with more advanced reference implementations
- Apply advanced imbalance-handling techniques such as SMOTE to further improve recall on rejected applicants
- Add authentication and role-based access for bank staff vs. self-service customer use
- Replace the local Flask development server with a production WSGI server (e.g., Gunicorn) behind a reverse proxy

Scalability Considerations

The current architecture (Flask + serialized model) can scale horizontally by deploying multiple instances behind a load balancer, and the model inference itself is lightweight enough for real-time use even at higher traffic volumes.